ANSWER KEY & MARKING SCHEME SET 1

Chapter 7: **Proportional Reasoning-1** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 63	One part $= 9$
2	**(c)** 24	Subtract the added number
3	**(c)** 18 L	$\frac{3}{10} \times 60$
4	**(c)** 15 hours	More pipes, less time
5	**(d)** 55	A : B : C = 4 : 6 : 5, so one part is 11
6	**(d)** 2 km	1 km = 100000 cm
7	**(b)** 10 m/s	Multiply by $\frac{5}{18}$
8	**(a)** 48	One part $= 6$
9	**(d)** 12	Subtract the added number
10	**(b)** 24 L	$\frac{2}{7} \times 84$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	One part $= 98 \div 14 = 7$ (1); numbers 35 and 63 (1).	1+1
12	45 km/h $= 45 \times \frac{1000}{3600}$ (1) $= 12.5$ m/s (1).	1+1
13	Actual distance $= 6 \times 50000$ cm $= 300000$ cm (1); $= 3$ km (1).	1+1
14	One part $= 90 \div 15 = 6$ (1); numbers 24 and 66 (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Milk 35 L, water 5 L (1); new water $= \frac{3}{7} \times 35 = 15$ L (1); add 10 L (1).	1+1+1
16	A : B : C $= 3 : 6 : 5$ (1); one part $= 10$ (1); numbers 30, 60, 50 (1).	1+1+1
17	One part $= 6$; daughter 18 (1); 4 years ago: 38 and 14 (1); ratio $19 : 7$ (1).	1+1+1
	OR: Milk 36 L, water 24 L (1); new water $= \frac{4}{3} \times 36 = 48$ L (1); add 24 L (1).	1+1+1
18	A : B : C $= 6 : 8 : 5$ (1); one part $= 7$ (1); numbers 42, 56, 35 (1).	1+1+1
	OR: One part $= 8$; daughter 16 (1); 6 years ago: 34 and 10 (1); ratio $17 : 5$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Total work $= 12 \times 15 = 180$ man-days (1); (i) $180 \div 18 = 10$ days (2); (ii) $180 \div 9 = 20$ men (2).	1+2+2
	OR: Ratio $3 : 5$ (1); one part ₹ 100 (1); Anil ₹ 300 (1.5); Bhanu ₹ 500 (1.5).	1+1+1½+1½
20	Total work $= 20 \times 24 = 480$ man-days (1); (i) $480 \div 16 = 30$ days (2); (ii) $480 \div 30 = 16$ men (2).	1+2+2
	OR: Ratio $2 : 3$ (1); one part ₹ 300 (1); Anil ₹ 600 (1.5); Bhanu ₹ 900 (1.5).	1+1+1½+1½

Mistakes to avoid:

- Forming a ratio without converting units: 50 cm : 2 m is $50 : 200 = 1 : 4$, not $25 : 1$.
- Writing the ratio in the wrong order: the ratio of A to B is not the same as B to A.
- Dividing a total in the ratio $3 : 5$ by dividing it by 3 and 5 separately; first add the parts ($3 + 5 = 8$) and divide by 8.
- Adding the same number to both terms to get an equivalent ratio ($2 : 3 \neq 4 : 5$); equivalent ratios are got by multiplying or dividing.
- Comparing ratios by only looking at the first terms; convert to fractions with the same denominator.
- Mixing up which quantities go together in the unitary method (e.g. dividing cost by the wrong number of items).

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

THE MASTERMIND COACHING CLASSES

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 7: **Proportional Reasoning-1** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 6 hours	More pipes, less time
2	**(b)** 45	A : B : C = 2 : 4 : 3, so one part is 15
3	**(d)** 6 km	1 km = 100000 cm
4	**(c)** 15 m/s	Multiply by $\frac{5}{18}$
5	**(a)** 70	One part $= 7$
6	**(c)** 20	Subtract the added number
7	**(d)** 50 L	$\frac{5}{9} \times 90$
8	**(c)** 4 hours	More pipes, less time
9	**(d)** 70	A : B : C = 6 : 8 : 7, so one part is 10
10	**(d)** 7 km	1 km = 100000 cm

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	63 km/h $= 63 \times \frac{1000}{3600}$ (1) $= 17.5$ m/s (1).	1+1
12	Actual distance $= 4 \times 200000$ cm $= 800000$ cm (1); $= 8$ km (1).	1+1
13	One part $= 75 \div 15 = 5$ (1); numbers 35 and 40 (1).	1+1
14	81 km/h $= 81 \times \frac{1000}{3600}$ (1) $= 22.5$ m/s (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Milk 20 L, water 25 L (1); new water $= \frac{8}{4} \times 20 = 40$ L (1); add 15 L (1).	1+1+1
16	A : B : C $= 6 : 9 : 4$ (1); one part $= 13$ (1); numbers 78, 117, 52 (1).	1+1+1
17	One part $= 5$; daughter 20 (1); 5 years ago: 40 and 15 (1); ratio $8 : 3$ (1).	1+1+1
	OR: Milk 45 L, water 27 L (1); new water $= \frac{6}{5} \times 45 = 54$ L (1); add 27 L (1).	1+1+1
18	A : B : C $= 3 : 9 : 7$ (1); one part $= 10$ (1); numbers 30, 90, 70 (1).	1+1+1
	OR: One part $= 4$; daughter 12 (1); 2 years ago: 30 and 10 (1); ratio $3 : 1$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Total work $= 9 \times 40 = 360$ man-days (1); (i) $360 \div 15 = 24$ days (2); (ii) $360 \div 12 = 30$ men (2).	1+2+2
	OR: Ratio $2 : 3$ (1); one part ₹ 400 (1); Anil ₹ 800 (1.5); Bhanu ₹ 1200 (1.5).	1+1+1½+1½
20	Total work $= 14 \times 18 = 252$ man-days (1); (i) $252 \div 21 = 12$ days (2); (ii) $252 \div 12 = 21$ men (2).	1+2+2
	OR: Ratio $2 : 3$ (1); one part ₹ 350 (1); Anil ₹ 700 (1.5); Bhanu ₹ 1050 (1.5).	1+1+1½+1½

Mistakes to avoid:

- Forming a ratio without converting units: 50 cm : 2 m is $50 : 200 = 1 : 4$, not $25 : 1$.
- Writing the ratio in the wrong order: the ratio of A to B is not the same as B to A.
- Dividing a total in the ratio $3 : 5$ by dividing it by 3 and 5 separately; first add the parts ($3 + 5 = 8$) and divide by 8.
- Adding the same number to both terms to get an equivalent ratio ($2 : 3 \neq 4 : 5$); equivalent ratios are got by multiplying or dividing.
- Comparing ratios by only looking at the first terms; convert to fractions with the same denominator.
- Mixing up which quantities go together in the unitary method (e.g. dividing cost by the wrong number of items).

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 7: **Proportional Reasoning-1** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 20 m/s	Multiply by $\frac{5}{18}$
2	**(d)** 44	One part $= 4$
3	**(b)** 35	Subtract the added number
4	**(c)** 42 L	$\frac{7}{12} \times 72$
5	**(a)** 12 hours	More pipes, less time
6	**(b)** 21	A : B : C = 4 : 10 : 3, so one part is 7
7	**(b)** 2 km	1 km = 100000 cm
8	**(a)** 25 m/s	Multiply by $\frac{5}{18}$
9	**(a)** 45	One part $= 5$
10	**(c)** 12	Subtract the added number

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Actual distance $= 9 \times 100000$ cm $= 900000$ cm (1); $= 9$ km (1).	1+1
12	One part $= 64 \div 16 = 4$ (1); numbers 12 and 52 (1).	1+1
13	126 km/h $= 126 \times \frac{1000}{3600}$ (1) $= 35$ m/s (1).	1+1
14	Actual distance $= 5 \times 250000$ cm $= 1250000$ cm (1); $= 12.5$ km (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Milk 30 L, water 36 L (1); new water $= \frac{9}{5} \times 30 = 54$ L (1); add 18 L (1).	1+1+1
16	A : B : C $= 8 : 10 : 3$ (1); one part $= 8$ (1); numbers 64, 80, 24 (1).	1+1+1
17	One part $= 7$; daughter 7 (1); 7 years ago: 35 and 0 (1); ratio $1 : 0$ (1).	1+1+1
	OR: Milk 60 L, water 20 L (1); new water $= \frac{2}{3} \times 60 = 40$ L (1); add 20 L (1).	1+1+1
18	A : B : C $= 4 : 10 : 9$ (1); one part $= 8$ (1); numbers 32, 80, 72 (1).	1+1+1
	OR: One part $= 9$; daughter 9 (1); 3 years ago: 33 and 6 (1); ratio $11 : 2$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Total work $= 16 \times 21 = 336$ man-days (1); (i) $336 \div 12 = 28$ days (2); (ii) $336 \div 14 = 24$ men (2).	1+2+2
	OR: Ratio $2 : 3$ (1); one part ₹ 240 (1); Anil ₹ 480 (1.5); Bhanu ₹ 720 (1.5).	1+1+1½+1½
20	Total work $= 25 \times 12 = 300$ man-days (1); (i) $300 \div 20 = 15$ days (2); (ii) $300 \div 15 = 20$ men (2).	1+2+2
	OR: Ratio $3 : 5$ (1); one part ₹ 120 (1); Anil ₹ 360 (1.5); Bhanu ₹ 600 (1.5).	1+1+1½+1½

Mistakes to avoid:

- Forming a ratio without converting units: 50 cm : 2 m is $50 : 200 = 1 : 4$, not $25 : 1$.
- Writing the ratio in the wrong order: the ratio of A to B is not the same as B to A.
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